

# Black Holes as Almost-Perfect All:All Entanglement Graphs: Interior Collapse, Horizon Wiring, and the Micro-Hole Phase Transition

Draft v0.7 — computational companion

`src/bh_graph` + `scripts/generate_figures.py` reproduce all figures (75 tests)

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## Abstract

We formalize the intuition that a black hole interior is an almost-perfect all:all entanglement graph. Internal edges cost no exterior space; each of the  $k$  exterior legs costs  $\sim l_p^2$  of horizon area, so  $A(k) = k l_p^2$  independent of interior node count  $N$ . We show (1) the complete graph  $K_N$  destroys interior distance and, under finite-speed random circuits, derives the Sekino–Susskind  $t_* \sim \log N$  fast-scrambling bound; (2) mass enters via a derived  $k^*(N) = 16\pi(\varepsilon N/l_p)^2$  fixed point, spin/charge consume part of the exterior budget (extremal Kerr keeps half the legs), and the  $k \rightarrow 0$  monogamy limit is a baby-universe pinch-off with an explicit nonlinear CKW frontier; (3) micro-holes are pointlike until  $k l_p^2 \geq 4\pi r_{\text{point}}^2$ , then pop a horizon via a computed QES/island crossing, with an LQG-style area gap. Evaporation as leg surgery reproduces the exact Page curve (Page 1993,  $-1/2$  nat correction included) with Haar-typical fluctuations, and Hayden–Preskill mirror recovery works past  $k \sim N/2$ . Simulations in `bh_graph` accompany each claim.

## 1 Sec 1: All:all = no interior space

Interior wiring  $G_N$  compared across  $K_N$ , chain, grid, random regular under SI spread (cover time = seed eccentricity). Only  $K_N$  has diameter 1 and cover time 1 at every  $N$  (Figs. 1–2). Finite-speed random-matching circuits (Appendix A) promote this to a derived  $t_* \approx \log_2 N / \log_2(1 + p)$ .

| family         | diameter         | $t_{\text{cover}}$ (SI) | $t_*$ (circuit) | gap              |
|----------------|------------------|-------------------------|-----------------|------------------|
| $K_N$          | 1                | 1                       | $\log_2 N$      | $N$              |
| chain          | $N-1$            | $\sim N/2$              | $\sim N/v$      | $\sim \pi^2/N^2$ |
| grid           | $\sim 2\sqrt{N}$ | $\sim 2\sqrt{N}$        | $O(\sqrt{N})$   | $O(1/N)$         |
| random-regular | $\sim \log N$    | $\sim \log N$           | $\sim \log N$   | $O(1)$           |

Table 1: Scrambling hierarchy; `bh_graph.scrambling`, `circuits`.

## 2 Sec 2: Horizon area counts exterior legs

Interior pairs  $N(N-1)/2$  plus  $k \ll N^2$  exterior legs:

$$A(k) = k l_p^2, \quad R(k) = \sqrt{k l_p^2 / 4\pi}, \quad (1)$$

Fig 1 — Sec 1: fast scrambling hierarchy



Figure 1: SI cover time and diameter: all:all stays 1.

so  $N$  drops out (Fig. 3). Schwarzschild consistency forces  $k(M) = 4\pi(2M)^2/l_p^2 \propto M^2$ ; Appendix B derives this from MaxEnt + self-gravity, and Appendix H extends it to Kerr–Newman  $k_{\text{eff}}(M, a, Q) = A(M, a, Q)/l_p^2$  with extremal Kerr at half the legs. Monogamy (linear toy Fig. 4; nonlinear CKW Fig. 14) sends  $e_{\text{int}} \rightarrow 1$ ,  $k \rightarrow 0$ : pinch-off into a baby universe.

### 3 Sec 3: Micro-hole phase transition

A point region of radius  $r_{\text{point}}$  embeds  $k$  legs iff  $k l_p^2 < 4\pi r_{\text{point}}^2$ . Hence

$$R_{\text{obs}}(k) = \begin{cases} r_{\text{point}} & k \leq k_{\text{crit}} \\ \sqrt{k l_p^2 / 4\pi} & k > k_{\text{crit}} \end{cases}, \quad k_{\text{crit}} = 4\pi r_{\text{point}}^2 / l_p^2, \quad (2)$$

with LQG-style quantization  $A \in \{0\} \cup [A_{\text{min}}, \infty)$  (Figs. 5–6). Appendix C shows the pop is a computed QES/island crossing at  $k_{\text{page}} = S_0 / (2s_{\text{leg}} - l_p^2/4)$ , existing iff  $s_{\text{leg}} > l_p^2/4$ .

## 4 Discussion

Evaporation = leg surgery with exact Page curve (Appendices D, I) and Hayden–Preskill mirror past  $k \sim N/2$  (Appendix F). Open: deriving  $\varepsilon$  and  $r_{\text{point}}$  coefficients from a UV embedding; Kerr Page curve with spin budget; OTOC Lyapunov check (Appendix L).

### A Finite-speed circuits: $t_* \sim \log N$ (Fig. 7)

One gate/qubit/step; random matching (all:all) vs dimer covering (chain). Early growth exponential on all:all; chain ballistic. Measured  $N = 8 \dots 128$ .

## B Deriving $k(N)$ (Fig. 8)

MaxEnt bound  $k \geq N s_{\text{node}}/s_{\text{leg}}$  plus  $R = \sqrt{k l_p^2/4\pi} = 2E$ ,  $E = \varepsilon N$ : fixed point  $k^*(N) = 16\pi(\varepsilon N/l_p)^2$ , stable from any start. Predicts  $\alpha(N) = k^*/N \propto N$ .

## C QES pop from entropy crossing (Fig. 9)

$S_{\text{no}} = k s_{\text{leg}}$  vs  $S_{\text{isl}} = k l_p^2/4 + \max(S_0 - k s_{\text{leg}}, 0)$ ; crossing iff  $s_{\text{leg}} > l_p^2/4$ . Explicit core+legs min-cut agrees.

## D Page curve from leg surgery (Fig. 10)

$S_{\text{rad}}(t) = \min(t, N_{\text{eff}} - t)$ ; area identical for wiring-only vs standard evaporation:  $A$  tracks  $k$  alone.

## E QEC mirror (Fig. 11)

$\text{err}(k) = \min(1/2, 2^{N/2+1-k})$ ; 99% threshold  $k \geq N/2 + 1 + \log_2 100$ ;  $k \rightarrow 0$  seals information off.

## F Robustness (Fig. 11)

Log slope  $1/\log_2(1+p)$  at all  $p$ ; exact quadraticity at all  $\varepsilon$ ; sharp QES boundary at  $s_{\text{leg}} = l_p^2/4$ .

## G Kerr–Newman (Fig. 12)

$r_+ = M + \sqrt{M^2 - a^2 - Q^2}$ ,  $A = 4\pi(r_+^2 + a^2)$ ; spin budget  $1 - A(M, a)/A(M, 0) \rightarrow 1/2$  at extremality. Area law survives.

## H Exact Page curve (Fig. 13)

$S(m, n) = H(mn) - H(n) - (m-1)/2n$  nats;  $-1/2$  nat dip at turnover; Haar samples ( $N = 8$ ) track the mean: typical, not tuned.

## I Nonlinear monogamy (Fig. 14)

Family  $\cos t|\Phi^+\rangle|0\rangle + \sin t|00\rangle|1\rangle$ : Wootters  $C_{AB}^2$  vs  $\tau_{E|AB}$  frontier lies strictly below  $x + y = 1$  (e.g. 0.70 at  $t = 0.3$ ); CKW  $\geq 0$  everywhere; baby universe at  $(1, 0)$ .

## J OTOC + phenomenology (Fig. 15)

$C(t) \sim e^{\lambda t}/N$  on all:all gives  $t^* = \log N/\lambda$  vs ballistic  $N/v$  on chains. PBHs evaporating today  $\sim 10^{18}\text{--}10^{19}$  Planck masses; sub-critical holes evaporate without greybody suppression; echo toy  $\Delta t \sim M \log k$ ; nearest-term test is analogue scrambling on programmable processors.

## K Tensor network derives $\varepsilon$ (Fig. 16)

Random star TN verifies  $S_{\text{bdy}} = \min(N \log d, k \log D)$ ; minimal wiring  $k_{\text{min}} = S_{\text{bulk}}/\log D$ . Two routes fix  $\varepsilon$ : QES coincidence  $\varepsilon = (l_p/N)\sqrt{k_{\text{page}}/16\pi}$  and crossover  $\varepsilon = l_p\sqrt{s_{\text{node}}/(16\pi \log D N_{\text{match}})}$ .

## L Kerr Page curve (Fig. 17)

$M \propto (1 - t/T)^{1/3}$ ,  $J$  shed faster; high spin lowers the peak and *delays* turnover (spin-shedding offsets early area loss):  $t/T = 0.73$  at  $a_0 = 0.95$  vs  $0.54$  Schwarzschild.

## M SYK ED vs chain (Fig. 18)

$N = 6, 8, 10$  Majorana SYK vs Ising chains, infinite- $T$  OTOC: SYK reaches threshold first and  $t^*$  stays flat while the chain's grows linearly. MSS saturation cited, hierarchy reproduced.

## N GWTC leg creation (Fig. 19)

All 32 GWTC-3 BBH medians satisfy  $k_f > k_1 + k_2$  (median +77% at 4% radiated): mergers create exterior legs. Spin caveat  $\sim 13\% \ll$  margin.

## O Hardware literature + prediction (Fig. 20)

Gärtner 2017 (all:all ions,  $m \leq 8$  OTOC coherences) and Mi 2021 (Sycamore grid, ballistic OTOC decay) both consistent. Head-to-head prediction at  $N = 53$ : 7.9 vs 14.6 steps; null result falsifies Sec 1/A.

## P TeV recast + echoes + EHT (Figs. 21–22)

LHC masses sit at  $k \sim 11\text{--}17$  vs  $k_{\text{crit}} \sim 50$ : thermal nulls expected. Echo spacings  $0.03\text{--}0.3\text{ s}$  land in searched windows (amplitude unmodeled). Shadow deviation  $\sim 10^{-48}$ : EHT-identical by 47 orders.

## Reproducibility

```
pip install -e .; python -m pytest tests/ -q; python scripts/generate_figures.py; streamlit run app.py.
```

## References

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